

Ex. No 5 Use Autopsy to create a case and import evidence

Description

Autopsy is an open-source digital forensics platform used for analyzing and extracting data from digital devices. Here's a step-by-step process on how to use Autopsy for a basic forensic investigation:

Drive link: https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v

Download files: [4Dell Latitude CPi.E01](#), [4Dell Latitude CPi.E02](#)

1. Installation

- **Download and Install:** Autopsy can be downloaded from the official website. Follow the installation instructions based on your operating system (Windows, Linux, or macOS).

2. Starting a New Case

- **Open Autopsy:** Launch the application after installation.
- **Create a New Case:**
 - Click on New Case.
 - Enter the case name and location where the case data will be stored.
 - Fill in the details like the case number, examiner's name, etc., and click Next.

3. Adding a Data Source

- **Choose the Type of Data Source:**
 - After creating a case, you'll be prompted to add a data source.
 - You can add disk images, directories, logical files, or local disks.
- **Select the Data Source:**
 - Browse to the location of the image file (e.g., .E01, .dd, or .raw), physical disk, or directory you want to analyze.
 - https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-v
- **Configure Ingest Modules:**
 - Autopsy allows you to select specific analysis modules such as File Type Identification, Keyword Search, Hash Lookup, etc.
 - You can enable or disable these based on your investigation needs.
- **Start Analysis:** Click Next to start the analysis.

1. Initial Analysis and Overview

- **Ingest Progress:** As Autopsy processes the data source, you'll see the progress in the lower-left corner.
- **Explore the Resulting Artifacts:**
 - Autopsy automatically categorizes findings such as web artifacts, file system metadata, and communication records.
- **Use the Tree Viewer:**
 - On the left pane, you'll see a tree structure where you can explore different aspects like File System, Web History, Email, etc.

5. Detailed Analysis

- **Keyword Search:**
 - You can perform specific keyword searches using the Keyword Search module.
 - Use pre-configured lists or enter custom keywords.
- **File Analysis:**
 - Navigate through files and folders under the File Types or File System section.
 - Open, view, or export files for further examination.
- **Timeline Analysis:**
 - Use the Timeline module to visualize events based on timestamps.
 - This can help track user activity over time.
- **Hash Analysis:**
 - Compare file hashes with known databases to identify known good or bad files.

6. Reporting

- **Generate a Report:**
 - After analyzing the data, click on Generate Report from the toolbar.
 - Choose the type of report (HTML, CSV, Excel, etc.).
 - Select which parts of the analysis you want to include in the report.
- **Export Findings:**
 - Export individual files or artifacts that you need for your report or further analysis.
- **Final Review:**
 - Review the report to ensure it includes all relevant information.
 - Save or print the report for use in your case.

7. Case Closure

- **Close the Case:**
 - Once you have completed your investigation, close the case within Autopsy.
- **Archiving:**

- Ensure all data and reports are properly archived according to your organization's policies.

8. Advanced Features (Optional)

- **Custom Ingest Modules:**
 - Autopsy allows for custom modules to be added if you need specific analysis tools not included by default.
- **Collaboration:**
 - Autopsy can be configured for multi-user cases if you're working in a team environment.

Opening new case:

The screenshot shows the 'New Case Information' dialog box. On the left, a 'Steps' sidebar lists '1. Case Information' and '2. Optional Information'. The main area, titled 'Case Information', contains the following fields and options:

- Case Name:** A text box containing 'EXP-5'.
- Base Directory:** A text box containing 'F:\', with a 'Browse' button to its right.
- Case Type:** Two radio buttons: 'Single-User' (selected) and 'Multi-User'.
- Case data will be stored in the following directory:** A text box containing 'F:\EXP-5'.

At the bottom of the dialog, there are five buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Opening new case:

The 'New Case Information' dialog box is shown with the 'Optional Information' tab selected. The 'Steps' list on the left indicates the current step is 'Optional Information'. The 'Case' section has a 'Number' field with the value '1'. The 'Examiner' section includes fields for 'Name' (B SHASHANK), 'Phone', 'Email' (99240041211@klu.ac.in), and 'Notes'. The 'Organization' section shows 'Organization analysis is being done for:' with a dropdown set to 'Not Specified' and a 'Manage Organizations' button. At the bottom are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

New Case Information

Steps

1. Case Information
2. **Optional Information**

Optional Information

Case

Number: 1

Examiner

Name: B SHASHANK

Phone:

Email: 99240041211@klu.ac.in

Notes:

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

Adding Data Source:

The 'Add Data Source' dialog box is shown with the 'Select Data Source' tab selected. The 'Steps' list on the left indicates the current step is 'Select Data Source'. The 'Path' field contains 'F:\4Dell Latitude CPLE01' with a 'Browse' button. There is an unchecked checkbox for 'Ignore orphan files in FAT file systems'. The 'Time zone' is set to '(GMT+5:30) Asia/Calcutta'. The 'Sector size' is set to 'Auto Detect'. The 'Bitlocker Password (optional)' field is empty. The 'Hash Values (optional)' section includes fields for 'MD5', 'SHA-1', and 'SHA-256', all of which are empty. A note at the bottom states: 'NOTE: These values will not be validated when the data source is added.' At the bottom are buttons for '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. **Select Data Source**
4. Configure Ingest
5. Add Data Source

Select Data Source

Path: F:\4Dell Latitude CPLE01 Browse

☐ Ignore orphan files in FAT file systems

Time zone: (GMT+5:30) Asia/Calcutta

Sector size: Auto Detect

Bitlocker Password (optional):

Hash Values (optional):

MD5:

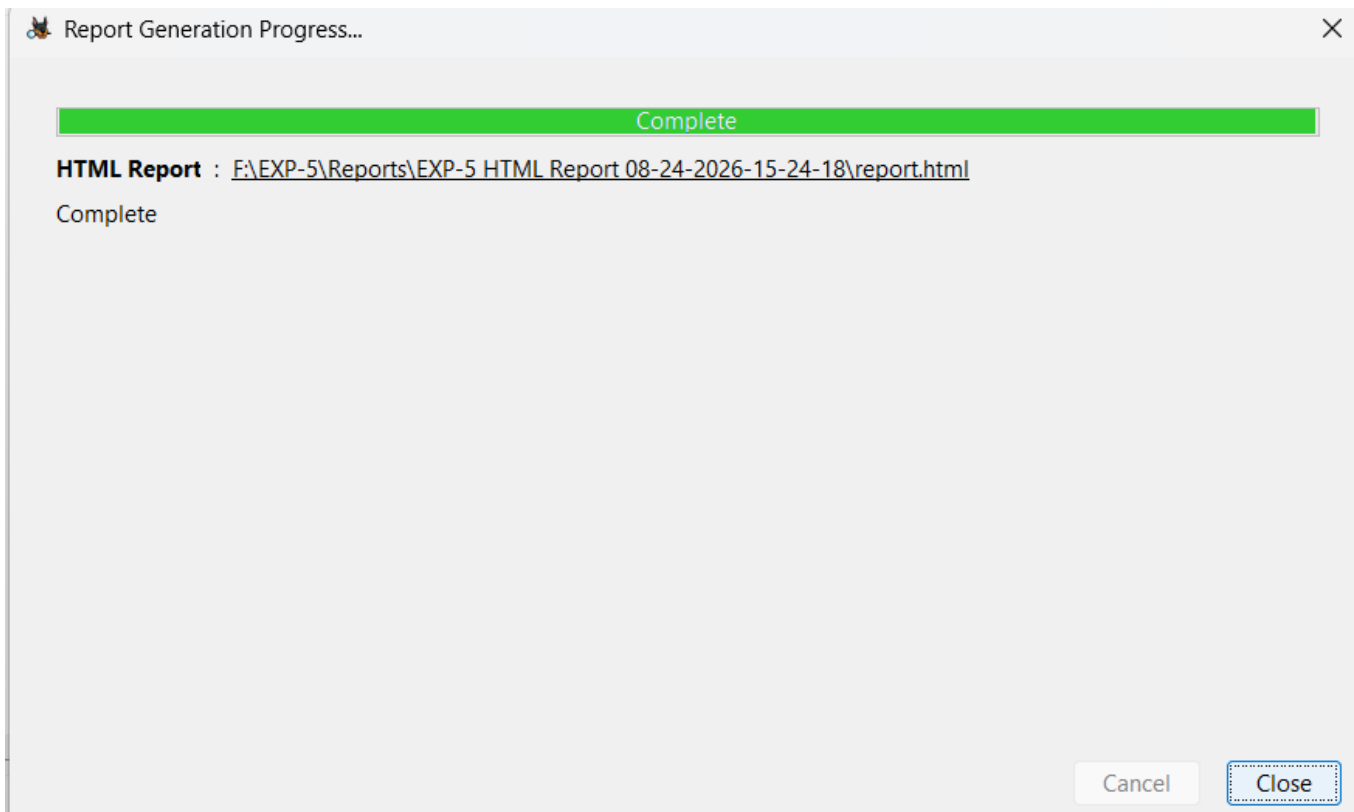
SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back Next > Finish Cancel Help

Generating Report:



Report:

Report Navigation

- Case Summary
- Data Source Usage (1)
- Encryption Suspicion (1)
- Extension Mismatch Detected (8)
- Interesting Items (4)
- Operating System Information (1)
- Recent Documents (8)
- Run Programs (28)
- Tagged Files (1)
- Tagged Images (1)
- Tagged Results (3)
- Web Bookmarks (5)
- Web Categories (3)
- Web Cookies (24)
- Web History (16)

Autopsy Forensic Report

Warning, this report was run before ingest services completed

2025-04-04 10:00:00 AM UTC-07:00

Case:	EXP-5
Case Number:	1
Number of data sources in case:	1
Examiner:	B SHASHANK

Image Information:

Image Location (Path):	
Timezone:	Asia/Calcutta
Path:	F:\001-Latitude-CP-EPT

Software Information:

Autopsy Version:	4.23.1
Android Analyzer Module:	4.23.1
Android Analyzer (dLEAPP) Module:	4.23.1
Central Repository Module:	4.23.1
DJL Drome Analyzer Module:	4.23.1
Data Source Integrity Module:	4.23.1
Email Parser Module:	4.23.1
Embedded File Extractor Module:	4.23.1
Encryption Detection Module:	4.23.1
Extension Mismatch Detector Module:	4.23.1
File Type Identification Module:	4.23.1
GPX Parser Module:	1.2
Hash Lookup Module:	4.23.1